

ALPARSLAN EROL

Principle Data Engineer



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Warsaw, Poland

Profiles:



PROFESSIONAL SUMMARY

Accomplished and results-driven Principal Data Engineer with over 7 years of experience in architecting, developing, and optimizing large-scale data pipelines and cloud-native data platforms. Proven leader in mentoring engineering teams, driving technical strategy, and aligning solutions with enterprise-level architectural patterns and implementing/leveraging AI into all DE processes. Expert in Databricks, Snowflake, Python, SQL, Airflow, Spark, dbt and building end-to-end data products on Azure and AWS. Passionate about leveraging cutting-edge technologies, including AI assistants, to enhance data quality, streamline processes, and deliver significant business impact. Recipient of the "H1 and H2 2025 D&AI Awards at PepsiCo".

SKILLS & CERTIFICATES

Data Engineering & Architecture: Python (Pandas, NumPy, Streamlit, Django, FAST API, Flask, Async Programming), SQL (Advanced Querying, Data Warehousing), dbt, ETL/ELT, Apache Airflow, Dimensional Modeling, Data Mesh, Data Contracts

Data Platforms & Warehousing: Databricks, Snowflake, Redshift, BigQuery, Spark, Hadoop, Hive, Kafka, InfluxDB

Cloud & DevOps Infrastructure: AWS (S3, EC2, Lambda, Glue, EMR EKS, ECS, Bedrock), Azure (AI Studio, AKS, Data Factory, Blob Storage, DevOps, AI-900), GCP (Cloud Storage, Dataflow, Cloud Run), Docker, Kubernetes, Terraform, Git, CI/CD

AI / ML, Analytics & BI: LLM, MCP, Transformer, LangChain, LangGraph, RAG, Embeddings & Semantic Search, Vector Databases, MLOPS, Feature Store, Regression, Classification, Deep Learning, Time Series, Power BI, Tableau, Grafana

PROFESSIONAL EXPERIENCE

PepsiCo

Remote

Principle Data Engineer – sDNA (Store DNA) Global

08.2025 – Present

- Serve as the technical lead for the sDNA data engineering team, providing mentorship to engineers, driving code quality through rigorous reviews, and establishing best practices for testing and automation.
- Architected and spearheaded the development of the next generation sDNA v3.0 data product, leveraging advanced Databricks features including Unity Catalog, Asset Bundles, and Lakeflow to enhance scalability and maintainability.
- Engineered fully automated, configurable data pipelines using Python, SQL, and Azure DevOps CI/CD, enabling seamless onboarding for over 20 countries into the sDNA ecosystem.
- Own the overall technical solution, ensuring alignment with enterprise-level architectural patterns while managing relationships and requirements with 10+ external data partners.
- Member of "AI Think Tank" initiative to scale the adoption of AI coding assistants across the D&AI organization, authoring blog posts on best practices and this initiative has nominated for a prestigious H2 2025 D&AI Awards.
- Implemented AI into all data engineering development processes (using MCP connections to read tasks from ADO-reading/writing documentations directly into Confluence-developing tasks with fully skilled Agentic AI workspace, democratizing data by using AI to BI design—users are able to get insights directly from data in Unity Catalog.

Senior Data Engineer – ROI Engine

10.2022 – 08.2025

- Engineered and scaled over 50 ETL pipelines using Airflow, Python, dbt and Snowflake, processing data from more than 20 distinct sources to drive data-driven business insights.
- Reduced new market onboarding time by 50% by developing and deploying automated global data ingestion and transformation pipelines for 10+ international markets.
- Decreased pipeline failures by 80% and achieved over 90% data accuracy by implementing more than 15 data quality monitors with Monte Carlo and Great Expectations.
- Contributed to the development of the award-winning ROI Engine, a project recognized with the H1 2025 D&AI Awards for its business impact and technical innovation.

Volvo Group

Göteborg, Sweden

Data Engineer – Under EY GDS

03.2022 – 11.2022

- Engineered robust ETL pipelines to process over 5TB of data per month, enabling a 25% reduction in truck breakdowns through predictive analytics.
- Optimized Azure Databricks Spark workloads, boosting processing speed by 60% and cutting infra costs by 30%.
- Centralized data platform architecture, reducing data redundancy by 40% and enhancing query performance.

EY GDS

Warsaw, Poland

Data Engineer – Consultant

11.2021 – 11.2022

- Working as Python Developer to support internal and external client-based projects with agile working methodology.
- Using cloud technologies mainly on Azure and AWS to design software back-end systems.
- Creating ETL pipelines for automatic Power-BI reporting tools with python, deploying the solution on serverless cloud solutions like Azure function or AWS Lambda, and storing data into Azure blob storage or AWS S3.
- Working as a full-stack developer by adding new features both in Django+Nginx on the backend with JavaScript+CSS.
- HTMX was used to build modern user interfaces with the simplicity and power of hypertext.

3M

Wroclaw, Poland

Data Engineer

8.2021 – 11.2021

- Creating data pipelines for mass sensor data from the production line by using Influx DB on the docker containers. Kubernetes is used to orchestrate the docker containers.
- Grafana tool is used to monitor the data stored in Influx DB.
- Store useful data on MongoDB for further analysis or Machine Learning models.
- Building some Machine Learning regression and Time Series Analysis models to predict the future outcome of the production line.

F-SECURE CORPORATION

Poznan, Poland

Junior Data Scientist - Artificial Intelligence Center of Excellence Team

09.2020 – 08.2021

- Working on data science R&D projects for the cyber security sector with Python backend, API, data engineering, and ML models development.
- Working fully agile by using Jira and version control system (GIT) with Bitbucket.
- Modelling and exploring the network data for the Detection and Response module.
- Using AWS S3 to store mass data for model training, ECS to docker cluster management about cost reduction project by using some windows, mac and Linux event filtering process, EMR on EC2 to manage big data platform clusters and control the model training logs, EC2 and Lambda to work on API.
- Experience in software development by processes/deploying on CI/CD, STG, and PRD layers.

DAIMLER TRUCKS / Mercedes-Benz Turkey

Istanbul, Turkey

ADAS Software Data Engineer - Daimler Truck R&D ADAS

09.2019 – 03.2020

- Developing Python backend software projects that pursued in Global Daimler R&D with Agile Method using Git.
- Developing a Deep Learning project based on LSTM (Long Short-Term Memory) to generate test data for functions in Advanced Driver Assistance System (ADAS) test and validation processes for these functions.

Bosch Turkey

Bursa, Turkey

Data Science Working Student

06.2019 – 09.2019

- Developed AI-Based Production Line Know-How Generator Project - a Machine Learning algorithm that uses a Gradient Boosting Method (LGBM) on Python environment.
- After two weeks of research about AI techniques for the production line, 80% of the total time was spent on "Data Preparation".
- Modeling, Training, Testing, and Evaluation processes done. For future ideas, "Fine Tuning" can be applied.

EDUCATION

University of Warsaw

Warsaw, Poland

M.Sc. in Data Science and Business Analytics, GPA: 4.6 / 5.0

08.2020-07.2022

Master Thesis: Investigating Graph Neural Networks on Traffic Signal Settings Optimization Problem

Polish Japanese Academy of Information Technology

Warsaw, Poland

M.Sc. in Computer Science, GPA: 4.6 / 5.0

02.2020-08.2020

Engineering of software, business processes, and databases specialized.

Boğaziçi University

Istanbul, Turkey

B.Sc. in Electrical & Electronics Engineering, GPA: 2.9 / 4.0

09.2016-01.2020

Specialization Areas: Control Systems, Optimization Theory, Robotics

Bachelor Thesis (Project): Network Anomaly Detection Using Machine Learning Algorithms / Python